

CANDIDATE
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INDEX NUMBER

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BIOLOGY

9744/03

Long Structured and Free response Question
Paper 3

14 September 2017
Thursday

Additional Materials: Answer Paper

2 hours

READ THESE INSTRUCTIONS FIRST

Write your name and PD group on all the work you hand in.
Write in dark blue or black pen.
You may use a soft pencil for any diagrams, graph or rough working.
Do not use paper clips, highlighters, glue or correction fluid.

Answer **all** questions.

All working for numerical answers must be shown.
At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [] at the end of each question or part question.

Calculators may be used

For Examiner's Use	
1	
2	
3	
4 OR	
5	
Total	55

Answer **all** the questions.

1

Cyanobacteria are a group of photosynthetic, nitrogen-fixing bacteria that live in a wide variety of moist soils and water either freely or in a symbiotic relationship with plants. Some cyanobacteria float in water by forming gas vesicles that are bounded by a protein sheath.

Fig. 1.1 below shows a generalized drawing of a cyanobacterium. The plasma membrane of cyanobacterium consists of an outer and inner membrane which is not represented in Fig. 1.1.

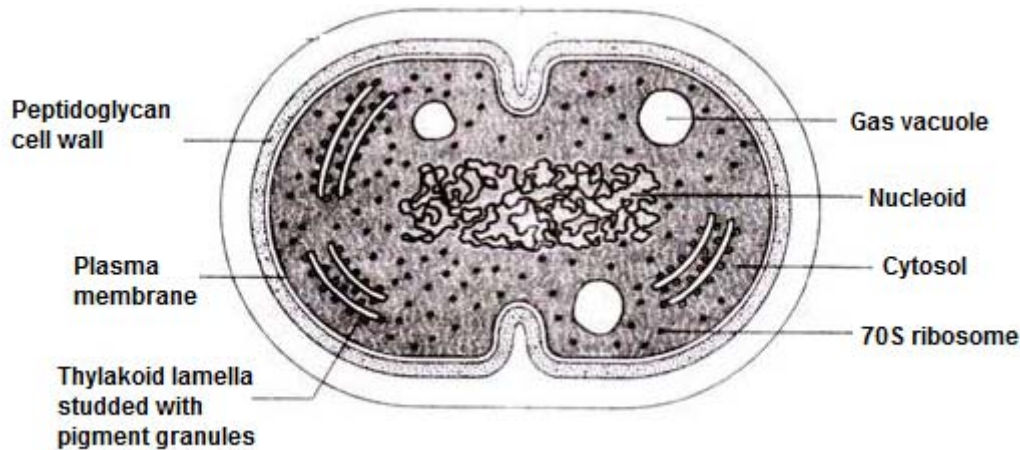


Fig. 1.1

(a) From Fig. 1.1, state two structural features that are expected in a typical prokaryote and two structural features that are not expected in a typical prokaryote.

Expected:

Not expected:

[2]

Expected: 70S ribosome, nucleoid/ DNA not bound by envelope, peptidoglycan cell wall (any 2)
Not expected: presence of membrane bound thylakoid lamella, gas vacuole
[1 mark for each]

(b) Process of photosynthesis that occurs in cyanobacterium is largely similar to photosynthesis in chloroplast. Fig. 1.2 shows the effect of carbon dioxide concentration on the light-independent stage of photosynthesis in *Synechococcus* genus of cyanobacterium. The following steps were carried out in a study:

- a cell suspension of *Synechococcus* was illuminated using a bench lamp.
- the suspension was supplied with carbon dioxide at a concentration of 1% for 200 seconds.
- the concentration of carbon dioxide was then reduced to 0.03% for a further 200 seconds.
- the concentration of RuBP and glycerate-3-phosphate (GP) were measured at regular intervals.
- the temperature of the suspension was maintained at 25 °C throughout the investigation.

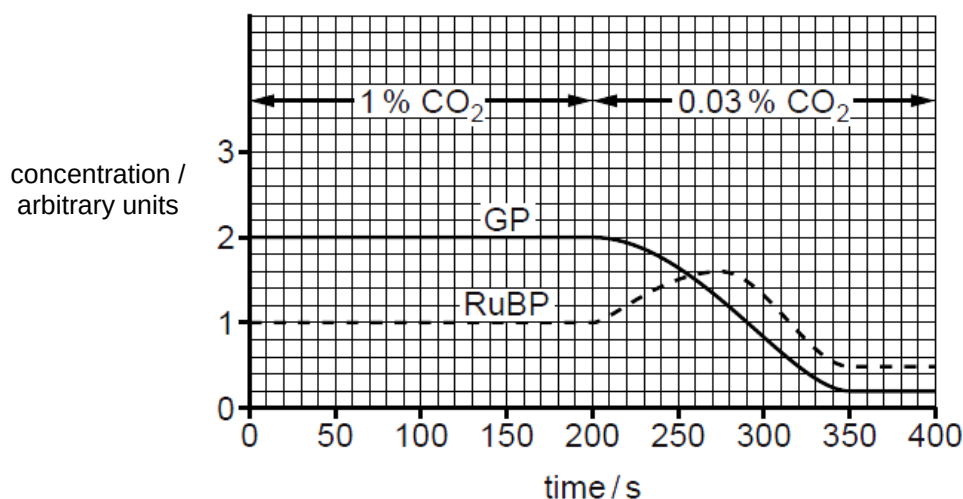


Fig. 1.2

(i) With reference to Fig. 1.2, explain why the concentration of RuBP changed between 200 and 275 seconds.

- concentration of RuBP increases from 1 a.u to 1.6 au when CO₂ concentration is lowered to 0.03%;
- decrease in frequency of effective collision between CO₂, RuBP and rubisco;
- rate of carbon fixation decreased, less RuBP is used/ RuBP accumulates;

[2]

(ii) Suggest how the decrease in the concentration of GP leads to an increase in the generation time (time it takes for the population to double) of *Synechococcus*.

- less glyceraldehyde-3-phosphate will be produced;
- so less conversion to carbohydrates / amino acids / proteins;
- less glucose for respiration to produce ATP for binary fission/ less amino acids to produce proteins for binary fission/ less raw materials for DNA binary fission;

[3]

(iii) Scientists have suggested that chloroplast may have originated as cyanobacterium that continued to function after becoming engulfed by primitive eukaryotic cells, in a process similar to endocytosis.

Describe two features of chloroplast that provide support for this hypothesis.

- presence of DNA/ ribosomes OR ability to synthesize proteins starting with DNA;
- DNA is circular;
- Idea of cyanobacterium is also double membrane like chloroplast/ idea of double membrane linked to endocytosis/ vesicle;
- Presence of 70S ribosomes;
- Multiply by binary fission;
- AVP

[2]

(c) The acquisition of photosynthetic bacterium would have provided the plants with nutritional independence, afforded by the ability to photosynthesize. It is hypothesized that these endosymbiotic associations were highly advantageous and thus naturally selected for in the course of evolution.

In the study of evolution of Man, the theory of natural selection is widely used to understand how speciation of humans has occurred. The study of fossils and genetic sequences are now commonly used to help us understand more about human evolution. It is widely believed that

humans are closely related to the Great Apes – chimpanzees, gorillas and orang utan, and share a common ancestor millions of years go.

Fig. 1.3 below shows some a comparison of skull structure between the Great Apes and modern Man (*Homo sapiens*).

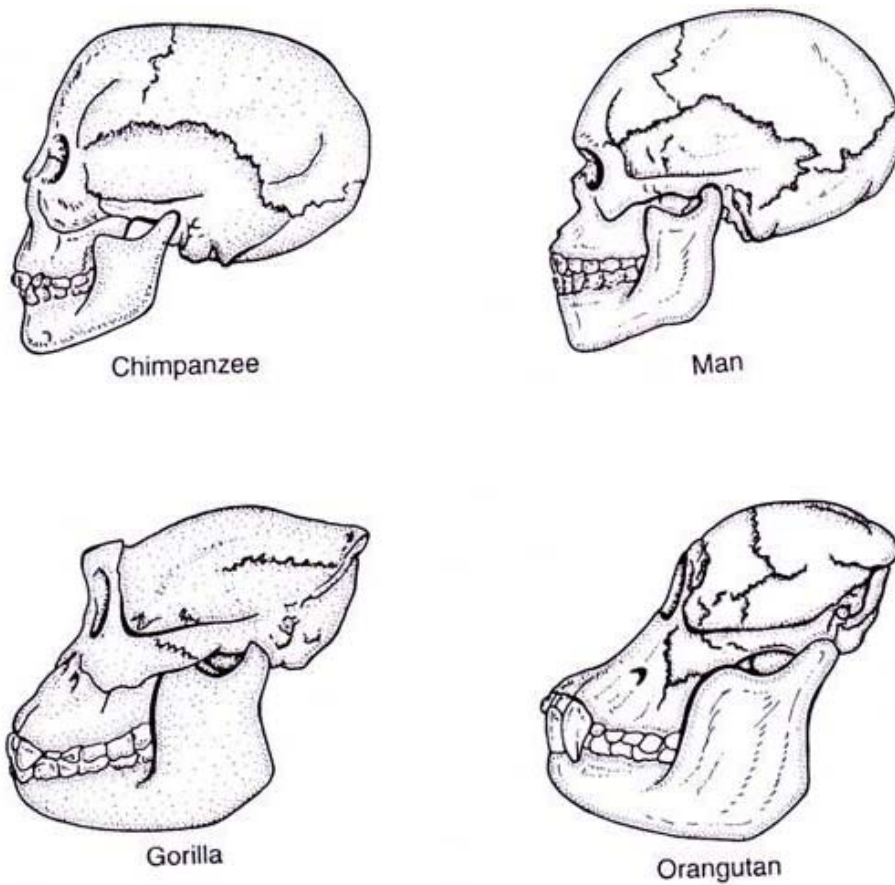


Fig. 1.3

(i) Using Fig. 1.3, state 2 features that support the hypothesis that modern Man shares a common ancestor with the Great Apes.

- Moveable lower jaw;
- Upper and lower sets of teeth;
- Position of cranial space/ brain same/ behind the eyes/ any other suitable description;
- Presence of eye sockets;
- AVP

[2]

Neanderthals (*Homo neanderthalensis*), another primate similar to modern humans is our closest human relative. Fig. 1.4 below shows a comparison between the fossilized skull of a Neanderthal and modern Man.

			Neanderthals Man  Fig. 1.4	
		(ii)	Disagreement exists as to whether the scientific name for Neanderthals should be <i>Homo sapiens</i> or <i>Homo neanderthalensis</i>. With reference to four different species concepts, explain why it is difficult to assign a scientific name to Neanderthals.	
			Any three from: • Biological species concept: cannot directly test whether humans and Neanderthals could interbreed to produce viable and fertile offsprings; • Ecological species concept: cannot determine if they share same niche/ idea that humans outcompeted Neanderthals therefore did share same niche; • Morphological species concept: incomplete knowledge of Neanderthal morphology from fossilized bones/ no easy way to quantify how much morphological changes defines a different species; • Genetic species concept: hard to quantify how much genetic difference defines a different species/ hard to get complete sequence data from Neanderthal; • Phylogenetic species concept: limited data to determine phylogeny/ hard to decide if split in phylogeny is at species level;	[4]
		(iii)	Discuss one advantage of using genetic sequences to study evolution of Man. • Analysis of molecular data is objective since differences in DNA/RNA/ amino acid sequences can be quantified and compared by analyzing nucleotide and amino acid sequences. • So that able to differentiate two fossils/ earlier species of man with similar morphologies based on molecular differences. • So that Scientists are able to use both living and dead specimen material in classification of organisms.	[2]
				[Total: 17]

- 2 B-lymphocytes respond to the presence of a non-self antigen by dividing as shown in Fig. 2.1.

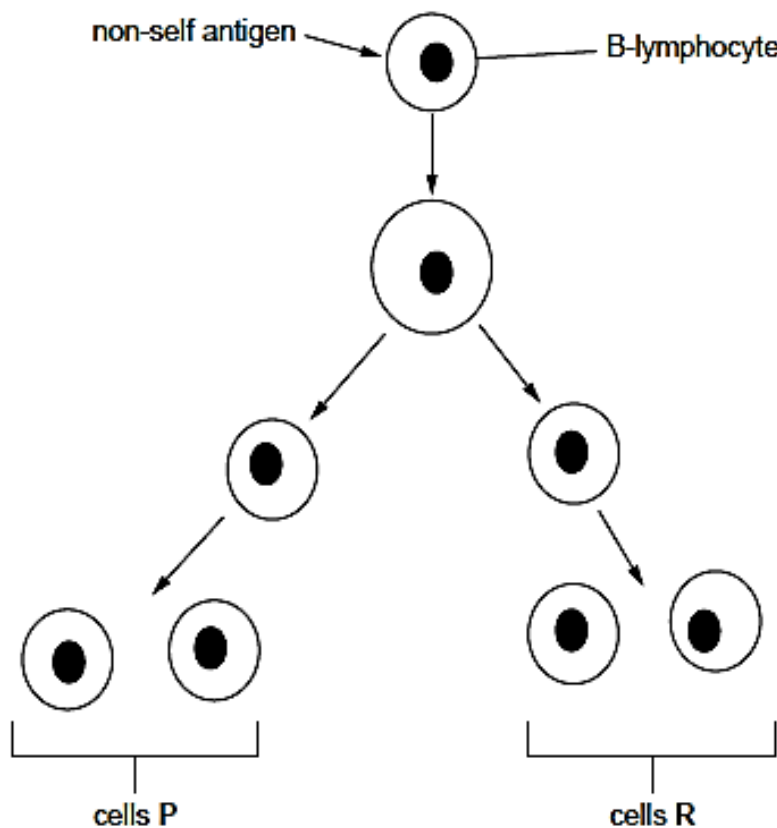


Fig. 2.1

- (a) During an immune response, cells divide by mitosis. Describe how mitosis is involved in an immune response.

1. Occurs in both primary and secondary immune response;
2. Clonal selection and expansion - Selected B and T lymphocytes divide by mitosis;
3. Mitosis in memory cells for rapid secondary response;

[3]

- (b) The cells labelled P on Fig. 2.1 continue to divide to give rise to many cells that differentiate into short-lived plasma cells. The plasma cells release antibody molecules.

- (i) Outline how plasma cells produce antibody molecules.

1. Using DNA as a template, gene is transcribed to mRNA;
2. Small ribosomal subunit binds to ribosome binding site at 5' end of mRNA, followed by attachment of large ribosomal subunit;
3. tRNA with complementary anticodon to mRNA codon carries specific amino acid to the ribosome;
4. formation of peptide bonds between adjacent amino acids;
5. synthesized polypeptide enters RER to fold into secondary and tertiary structure;
6. vesicle containing polypeptide buds off from RER and fuse with Golgi body;
7. undergoes further modification such as glycosylation/ formation of quaternary structure and disulphide bonds in Golgi apparatus;

[4]

- (ii) Describe how antibody molecules are released from the plasma cell.

Vesicles migrate/move and fuse with to cell surface membrane;
Release antibodies via exocytosis;
Movement of vesicle/exocytosis requires ATP;

[2]

- (c) Both B and T lymphocytes are part of adaptive immunity. Describe the mode of action of T-

		lymphocytes during an immune response.	
		T cells are activated when they encounter antigen in association with MHC II on another APC eg. macrophage; Those T cells that have receptors complementary to the antigen respond by dividing by mitosis and undergo clonal selection and clonal expansion to produce clones of T cells; T helper secrete cytokines to activate B- lymphocytes to divide and differentiate into plasma cells which will secrete antibody; Increase antibody levels for agglutination, neutralisation, opsonisation, ADCC, complement activation, immobilization of bacteria (any 2) Some T helper cells secrete cytokines that stimulate macrophages to carry out phagocytosis more vigorously; Some helper T cells secrete cytokines that stimulate killer T cells to divide by mitosis and to differentiate by producing vacuoles full of toxins; Cytotoxic T cells attach to kill infected cells by release of cytotoxic substances; Or by releasing perforin which will induce pores in the infected cell surface membrane for cell lysis/ granzymes which will trigger cell apoptosis; Memory helper T cells and memory killer T cells are produced, which remain in the body and become active very quickly during the secondary response to antigens; Natural killer (NK) cells - The F_{ab} portion of IgG binds with the target cell (microorganism or tumour cell) and the F_c portion binds with specific F_c receptors (<i>structure</i>) that are found on natural killer (NK) cells. NK cells destroy the target by releasing toxic substances contained in its cytoplasm granules and not by phagocytosis	[4]
	(d)	Immune response is mounted against pathogen such as bacteria. Explain why phagocytes act only against the bacteria and not against human cells.	
		1. bacterial's antigens are non-self/foreign and human cells have self antigens; 2. non-self and self antigens are proteins of different amino acid sequence/ self antigens are encoded by genes in the body; 3. non-self antigen will trigger phagocytosis by APCs (macrophages and dendritic cells); 4. phagocytes bind to antibodies complexed with non-self antigens/ human cells will not have bound antibody;	[4]
			[Total: 17]

- 3** The olive tree, *Olea europaea*, is a small tree native to the Mediterranean area of Europe, Africa and parts of Asia, where it has been cultivated for several thousand years. In 1993, Beerling and Chaloner carried out estimates of stomatal density on preserved olive leaves. The oldest of these were obtained from the tomb of the Egyptian King Tutankhamun who died over 3000 years ago. The results of the study are summarised in Fig. 3.1.

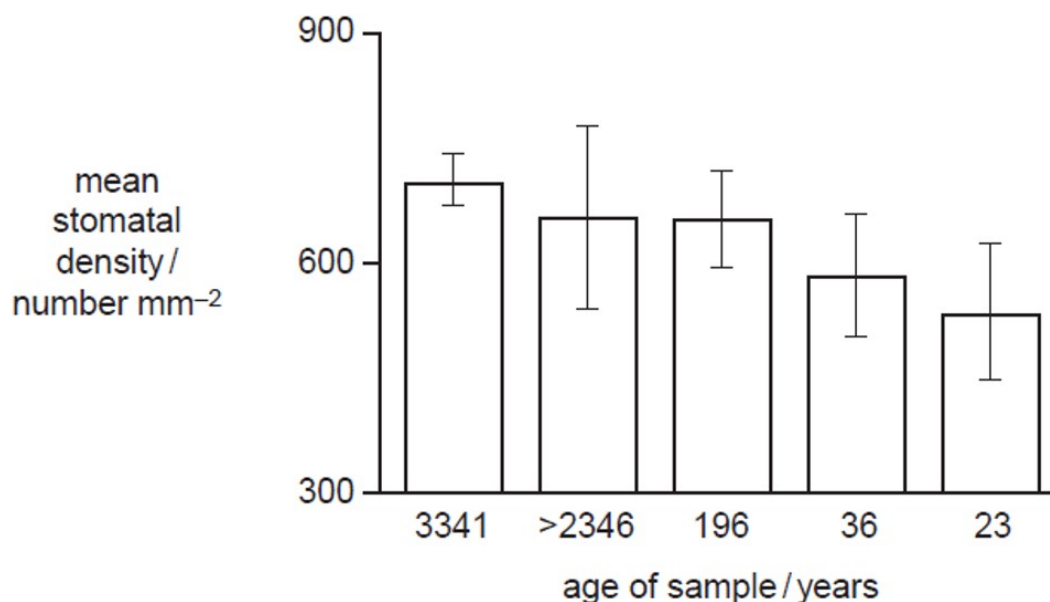


Fig. 3.1

(a)	(i)	Describe the results shown by the data in Fig. 3.1.	[2]
		- mean stomatal density decreases (cite data); - stomatal density has decreased steeply since 196 BP ; - no change between >2346 BP and 196 BP ; (any 2)	
	(ii)	Explain why it is difficult to reach a valid conclusion about changes in stomatal density over time	[4]
		- error bars show variation within samples / large variation within samples ; - overlapping errors bars indicates no significant difference between (most) means / samples ; - error bars for 3341 BP and 23 BP do not overlap ; - there is a significant difference over, 3318 years / period of study ; - a very long period / approximately 3500 years, represented by only five means ; - second sample, could be any age from 2346 BP / might be older than the first sample ; - large periods of time between samples / changes could have taken place between sampling dates ; - historical samples, are / likely to be, very small / non-representative ; - Any valid point; (Any 4)	

Over the last 10 years, Kenya has made progress in malaria control. However, the country is still far from defeating the disease.

Fig. 3.2 shows how prevalence of malaria is across the country.

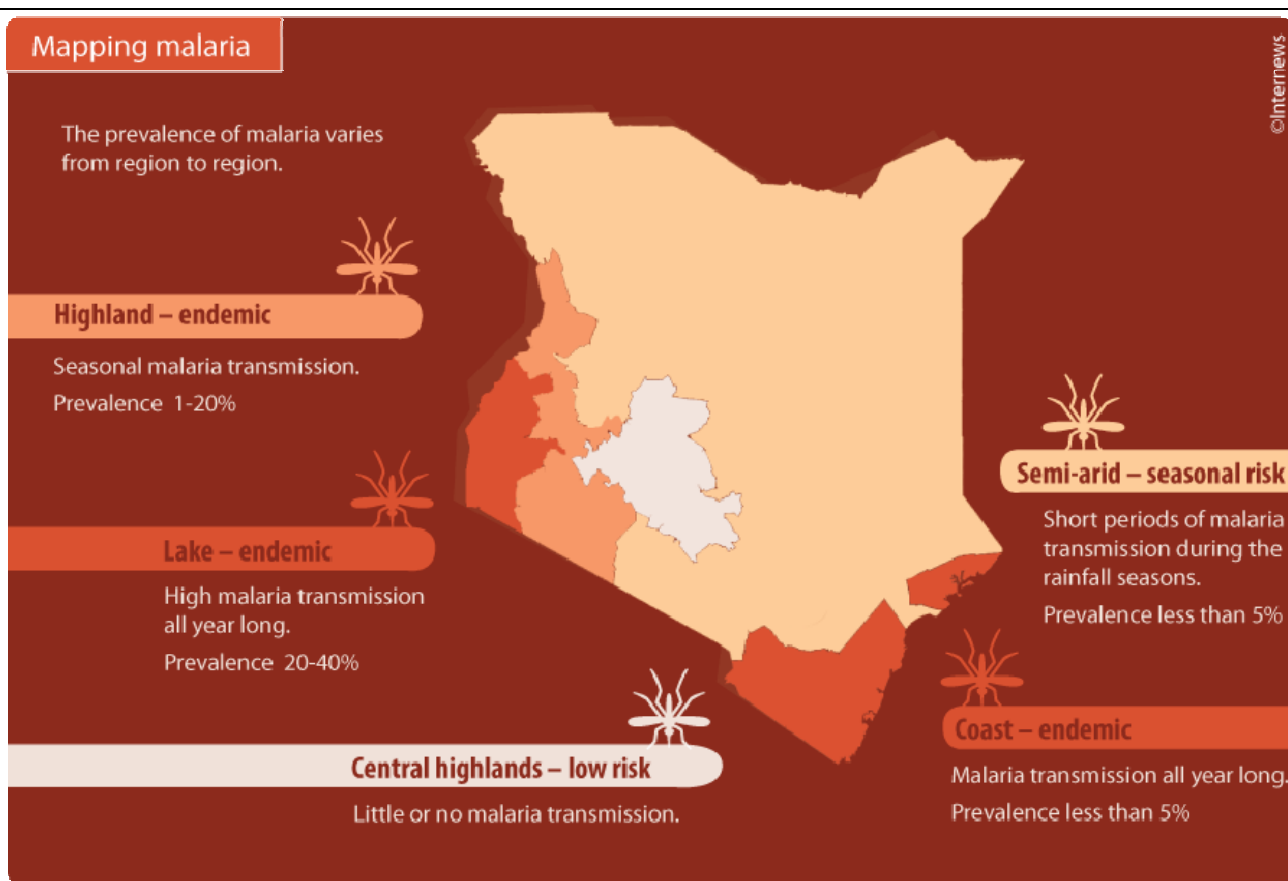


Fig. 3.2

(b)	(i)	With reference to Fig. 3.2, explain the two determining factors that lead to uneven prevalence of malaria across the country.	[4]
		Water - Higher prevalence near lake area (20-40%)/ Semi-arid areas only have seasonal risk during the rainfall seasons. Hence prevalence is less than 5% - Due to breeding area for mosquitoes; Temperature - Highland or central highland has less % prevalence than coast (all year long) - Due to warm temperature needed to complete the mosquito life cycle – state 1 e.g. egg laying; or metabolism	
	(ii)	Suggest why it is difficult to control malaria worldwide, apart from reasons associated with global warming.	[2]
		- Poor awareness of how mosquito spreads/ how to prevent being bitten/ effective government programme; - Resistance of mosquito to insecticide; - Resistance of plasmodium parasite to antibiotics; - Any valid reasons (Any 2)	
	(c)	Besides mosquito-borne diseases, describe two other problems caused by a change in insect population as a result of climate change.	[4]
		- Increase in named insect attacks on named crop; - affects crop yield which affects food security/ cattle feed; Migration of insects to higher altitude; - Affects food chains/ food webs of ecosystem;	
[Total: 16]			

Free-response question

Write your answers to this question on the separate answer paper provided.

Your answer:

- should be illustrated by large, clearly labelled diagrams, where appropriate;
- must be in continuous prose, where appropriate;
- must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

4	(a)	Describe the production and folding of a functional enzymatic protein that is used within a cell.	[12]
		Describe transcription: (max 5m) • Ref. to transcription of gene on template strand • Ref. to RNA polymerase binding to promoter region • Ref. to RNA polymerase adding free ribonucleotide triphosphates and are joined together by phosphodiester bonds • Ref. to elongation of mRNA in a 5' to 3' direction • Ref. to transcription stops after RNA polymerase reads the termination signal (in prokaryotes) / polyadenylation signal (in eukaryotes) • Ref. to pre-mRNA undergoes post-transcriptional processing of 5'capping, splicing and poly-A tailing (any 1 e.g.) to form mature mRNA; Describe translation: (max 5m) • mature mRNA translocates from nucleus to cytoplasm through nuclear pores; • Ref. to initiation of translation at 5' end of mRNA by assembling small and large ribosome and initiator tRNA; • Ref. to tRNA with complementary anticodon forms complementary base pairs with codon sequences, bringing specific amino acids to ribosome; • Ref. to formation of peptide bonds catalysed by peptidyltransferase in large ribosomal subunit; • Ref to stop codon and addition of release factor to A site; • Ref. to hydrolysis of ester bond between amino acid and tRNA, releasing polypeptide chain from ribosome; Describe protein folding: (max 5 m) • Enzymes are <u>globular</u> proteins with unique three-dimensional conformation / tertiary / quaternary structure; • Ref to primary structure being the unique <u>sequence and number of amino acids</u> in a polypeptide linked by <u>peptide bonds</u>; • Ref to secondary structure being the <u>regular</u> coiling and folding/pleating of the polypeptide held by <u>hydrogen bonds</u> between <u>CO and NH groups</u> of the <u>peptide bonds / polypeptide backbone</u>; • In <u>alpha helix</u>, <u>hydrogen bonds</u> form between CO and NH groups <u>4 a.a. apart</u>, forming a 3D <u>helical</u> structure • In <u>beta pleated sheet</u>, <u>hydrogen bonds</u> form between CO (or NH) group of one region/segment and NH (or CO) group of an <u>adjacent region/segment</u> of a <u>single polypeptide</u> chain, forming a <u>flat/pleated sheet</u>; • Tertiary structure refers to the folding of polypeptide into a specific conformation, held by <u>bonds between R-groups*</u> of structural amino acids <u>within same polypeptide</u>, maintained by <u>hydrophobic interaction, hydrogen bonds, ionic bonds, disulfide bridges</u>; • Ref to quaternary structure: <u>more than 1 polypeptide chain</u> to form functional protein held by <u>hydrophobic interaction, hydrogen bonds, ionic bonds, disulfide bridges</u> between R groups between polypeptide chains; • Folding gives rise to a specific cleft / groove - <u>active site</u> that is <u>complementary in shape and charge</u> to its <u>substrate</u>.	

- Folding brings catalytic amino acids and binding amino acids far apart in the primary structure / polypeptide close together in the active site

Overall max 12m.

(b) Discuss the importance of anaerobic respiration, and why it produces few ATP.

[13]

Importance of anaerobic respiration (max 4)

- Produces ATP even though no oxygen is available;
- Ref to heart pumping at maximum rate, and not delivering enough oxygen to mammalian muscle
- Importance to human: Yeast produce alcohol for use.
- Reference to death/ ATP used for cellular processes/ allow ATP to be made, so few ATP is better than none.
- AVP;

Why it produces few ATP (max 10)

- No oxygen as final electron acceptor;
- No oxidative phosphorylation, no Krebs cycle and no link reaction;
- Only glycolysis takes place;
- Produces 2 ATP per glucose;
- By Substrate level phosphorylation;
- 19 times lesser compared to aerobic respiration/ aerobic respiration makes 38 ATP per glucose;
- Due to NAD and FAD not regenerated by Oxidative phosphorylation;
- Aerobic respiration produces a lot of ATP because 1 NADH and 1 FADH₂ give 3 ATP and 2 ATP respectively per glucose.
- However, NAD is regenerated by fermentation/ NADH made in glycolysis is regenerated to NAD in fermentation;
- Energy is still trapped in ethanol and lactate;

[Total: 25]

5 (a) Discuss the importance of membranes in the reproductive cycle of the influenza virus.

[12]

1. Envelope/ cell membrane of viruses contains haemagglutinin (HA/H); which mediates the binding of the virus to specific receptor sites containing sialic acid sugars on the cell surface membrane of epithelial cells;; (especially in the nose, throat and lungs of mammals and intestines of birds);
2. Importance: allows the virus to recognize and attach to specific host cells;
3. The virus enters by endocytosis as the host cell membrane invaginates, forming an endocytic vesicle / endosome;
4. Importance of membrane: fluid nature of the membrane allows formation of vesicles;
5. Within the endosome, the acidic pH causes the hemagglutinin protein to undergo a conformational change;
6. Importance of membrane: allow the setting up of an acidic medium within the endosome (with the help of proton pumps on endosome membrane);
7. Resulting in the fusion of the viral envelope with the endosome membrane;
8. releasing the nucleocapsid into the cytoplasm;
9. (During synthesis of the new viruses) glycoproteins(HA and NA) are first synthesized in the rER; and then chemically modified in the Golgi apparatus (both are single membrane bound organelles in the host cells);;
10. The glycoproteins are then transported to the cell membrane via (secretory) vesicles (pinched off the Golgi apparatus);
11. These vesicles then **fuse** with the host cell membrane, thereby incorporating/**embedding** the glycoproteins into the (host) cell surface membrane;
12. These sites then serve as exit point for viral release;

13. New viruses leave the host cells through **budding**; thereby **acquiring the host cell membrane (=envelope)**;
14. Neuraminidase (NA/N), present on cell membrane helps in the release of new viruses by infected cells by cleaving off the sialic acids present on the host cells;

(b) Distinguish the differences in transcriptional control between *E. coli* and yeast cell, and explain the significance of post-translational control in yeast cell.

[13]

Point of comparison	<i>E. coli</i>	Yeast cell
Operon	Allows bacteria to coordinately regulate a group of genes that encode gene products with <u>related</u> functions.	Group of genes that encode gene products with <u>related</u> functions are located on different chromosomes.
Regulatory gene – repressor and binding site	Codes for a repressor which binds to operator	repressor protein binds to silencer (distal control element)
activator and binding site	Catabolite Activator Protein (CAP) binds to CAP binding site on promoter.	Activator protein binds to enhancer (distal control element)
Promoter	One promoter controlling a group of structural genes coding for enzymes involved in the same metabolic pathway	One promoter for each gene
mRNA	Polycistronic mRNA one mRNA code for more than 1 protein Presence of several start and stop codons	Monocistronic mRNA one mRNA code for 1 protein 1 start codon 1 stop codon
Accessibility of RNA polymerase to promoter	Inducible – inducer binds to active repressor allosterically to change its conformation such that it becomes inactive as its binding site is no longer complementary to operator sequence, therefore RNA polymerase is able to bind to promoter/ repressible operon – co-repressor (trp) binds allosterically to inactive repressor to change its conformation such that it becomes active as its binding site is now complementary to operator sequence, therefore RNA polymerase is no longer able to bind to promoter.	State at least two: Histone Acetylation / Histone methylation/ DNA methylation
Number of chromosomes involved	Operon is within the bacterial chromosome	More than one chromosome involved resulting in a coordinated response.

(8 maximum from above)

Importance of post-translational control:

- a) Cleavage and/or covalent modification
Give 1 suitable example - glycosylation, disulfide bond formation, attachment of prosthetic groups etc. is required.
- b) Form functional protein - newly synthesized proteins need to be modified for proper assembly / functioning
- c) Regulate - control cellular activity / influence biological activity
- d) Eg: phosphorylation/dephosphorylation may activate/inactivate proteins (e.g. kinases in phosphorylation cascade)
- e) Degrade proteins allows control of protein activities OR prevent aberrant activities so that proteins will not stay too long in cytoplasm and still be active
- f) E.g. Proteins are linked to ubiquitin that will target a protein for degradation.
- g) (Save/recycle resources) - proteins not needed can be hydrolysed to amino acids, to be used for synthesis of new proteins
- h) (Heterogeneity) - many different proteins modified from one polypeptide serve different function so smaller no of proteins/ small genome needed
- i) (Localisation) – direct proteins to particular locations inside and outside cell
- j) Eg: modifications at terminus of amino acid chain help target proteins for transporting to final destination in the cell / move across membranes/ tag proteins to be incorporated in various cellular and organelle membranes

(7 maximum from above)

[Total: 25]